

Representational Geometry in Neural Networks

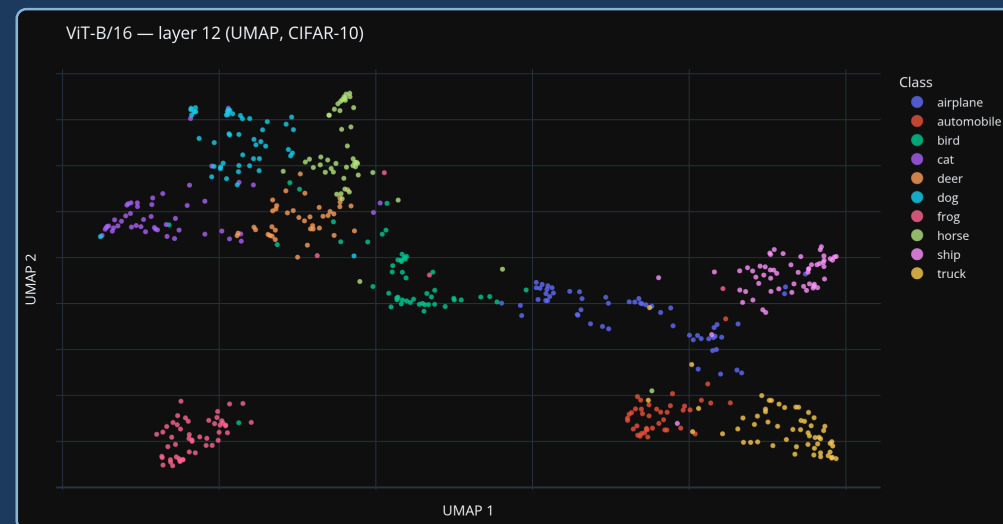
From Vision Transformers to RLHF-Aligned Language Models

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ViT-B/16 · layer 12 · CIFAR-10 · real data

Goal & Structure

What We Are Studying

Core question: Does RLHF alignment change the geometry of language model representations — and does it bring them closer to how vision models encode the visual world?

Part 1 — Visual geometry

ResNet-18 vs. ViT-B/16 on CIFAR-10
How do CNN and ViT representations differ across layers?

CKA + Class Activation Maps

Part 2 — Language geometry

Llama-3-8B base vs. Llama-3-8B-Instruct
How does RLHF reshape the embedding space?

LinearSVC + UMAP · Anthropic/hh-rlhf

Bridge → Matched (image, caption) pairs from MS-COCO let us measure **cross-modal CKA**. CLIP sets the ceiling (explicitly trained for this) — does RLHF move Llama toward it?

Approach

Part	What we measure	How
Vision	Layer-wise representational similarity between CNN and ViT	CKA — score in $[0, 1]$, invariant to rotation & scaling
Vision	Which image regions activate each architecture's layers	GradCAM — gradient-weighted spatial attention maps
Language	Linear separability of chosen vs. rejected response embeddings	LinearSVC per layer — accuracy & margin across 32 layers
Language	Geometric effect of RLHF alignment across the full model	UMAP scatter — base vs. instruct, layer slider
Cross-modal	Alignment between vision and language representations	CKA on matched MS-COCO (image, caption) pairs
Cross-modal	Does RLHF shift language geometry toward visual representations?	Llama-base vs. Instruct vs. CLIP (upper-bound baseline)

Current State

What We Have Built

Done

- Dash app · CKA heatmap, CAM viewer & layer UMAP
- Real CIFAR-10 embeddings · ResNet-18 ✓, ViT training
- Real LLM embeddings (Llama-3-8B base + Instruct)
- CKA + LinearSVC + UMAP pipeline · 24 tests

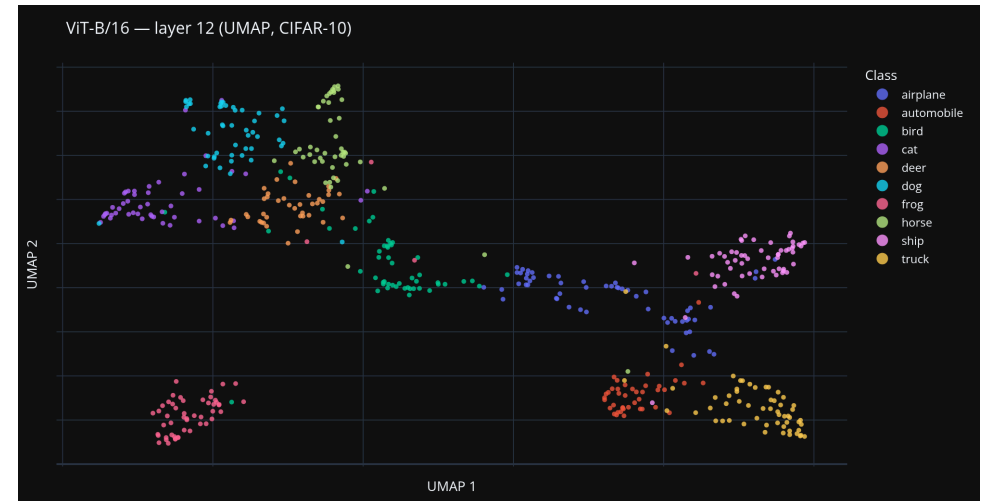
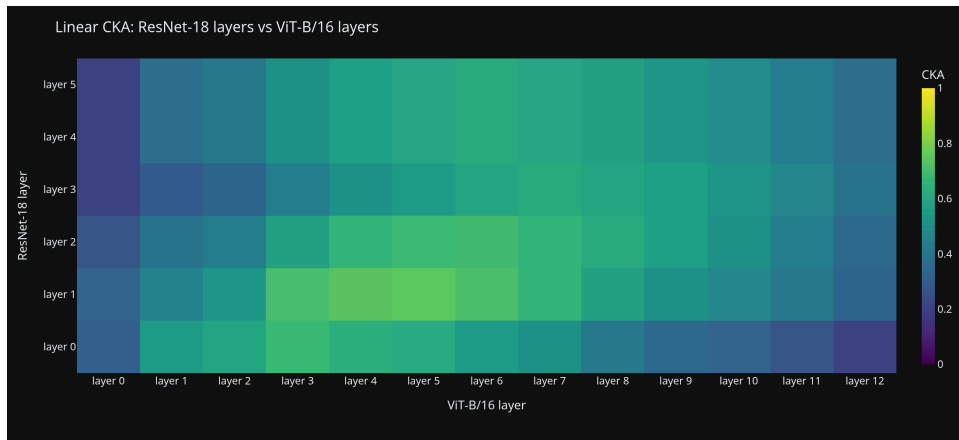
Up next

- ViT fine-tuning $\rightarrow \geq 85\%$ on CIFAR-10
- Cross-modal CKA: Llama-base / Instruct / **CLIP** × ViT
- Does RLHF move Llama toward CLIP-level alignment?
- Bias probe experiments (Part 2)

Early finding — ViT-B/16 builds class-discriminative structure monotonically across layers (UMAP ratio $0.44 \rightarrow 3.79$). ResNet without fine-tuning shows no class clustering — motivates proper training.

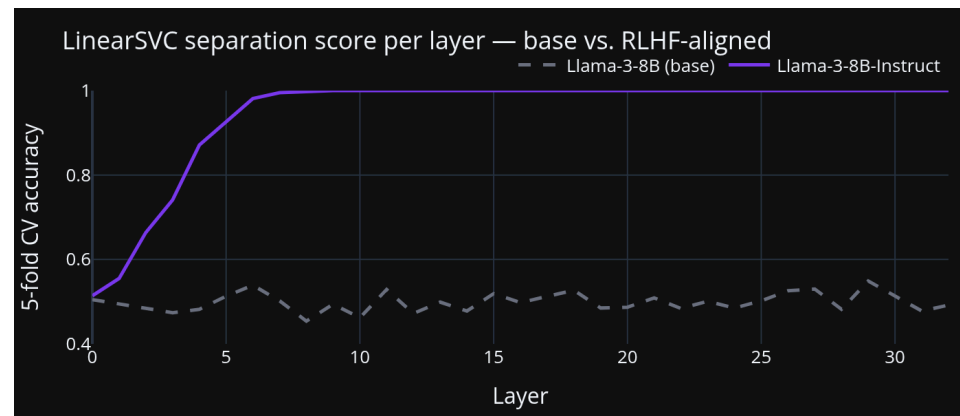
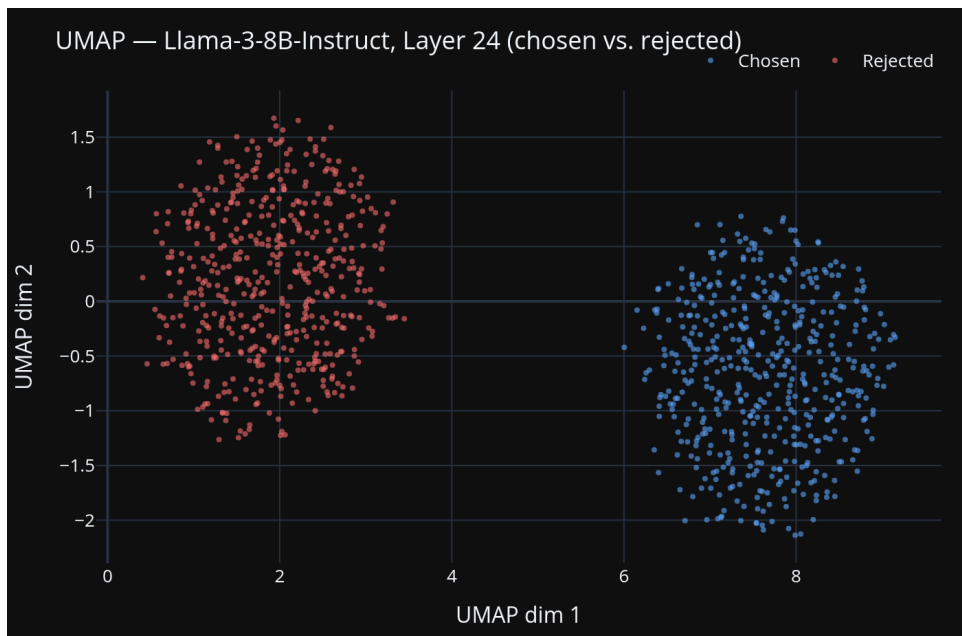
Dashboard

Part 1 — CKA Heatmap & Layer Activation UMAP



ViT-B/16 · layer 12 · CIFAR-10 test set (real data)

Part 2 — RLHF Embedding Space & Separation Score



Key finding so far

The aligned model shows markedly higher SVC accuracy in deeper layers — RLHF **geometrically** encodes human preference as a linear structure.